

ISTANBUL MEDİPOL UNIVERSITY
School of Engineering and Natural Sciences
2022- 2023 Spring



PROJECT MANAGEMENT
Final Project Report

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June 11

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Abstract

Inventory Software Project aims to create a smart inventory management system using advanced technologies. With the use of technologies such as IoT (Internet of Things), this system will provide real-time tracking of inventory and provide complete visibility of inventory. The positions, quantities and movements of the products will be monitored through sensors and labels, and this information will be instantly recorded in the system.

In addition, this smart inventory management system will automate the stock replenishment process. Inventory levels at points of sale will be monitored and when it falls below a certain threshold, the system will automatically notify suppliers and initiate the restocking process. In this way, errors and delays related to manual inventory tracking and stock control will be avoided, and customer requests will be responded to faster.

The project also intends to optimize inventory levels using forecast and historical sales data. This will ensure that inventory is kept at the right levels by making demand forecasts and trend analysis. In this way, overstock or shortage problems will be reduced and businesses will be able to better manage costs.

In summary, this project aims to provide real-time tracking of inventory, automate the stock replenishment process, and optimize inventory levels by creating a smart inventory management system using advanced technologies. In this way, significant improvements will be achieved in accuracy, efficiency and customer experience, and inventory management of enterprises will be carried out more effectively.

Introduction

The purpose of the final project is to develop a smart inventory management system that will revolutionize the way businesses manage their inventory and streamline their operations. Efficient inventory management plays an important role in the success of any business, as it directly impacts profitability, customer satisfaction, and overall efficiency. By leveraging cutting-edge technologies such as the Internet of Things (IoT), this system aims to provide businesses with real-time visibility, control, and optimization of their inventory.

The scope of this project encompasses the design, development, and implementation of a robust and intelligent inventory management system. The system will utilize IoT technology to track and monitor inventory items throughout the supply chain, from procurement to storage, sales, and distribution. By connecting physical inventory with digital systems, businesses will gain unprecedented insights into their inventory levels, location, and movement, allowing for more accurate demand forecasting, efficient replenishment, and enhanced decision-making.

The smart inventory management system assumes the availability of a reliable internet connection to ensure seamless communication between IoT devices and the central inventory management platform. Furthermore, it assumes the system's integration capability with existing enterprise resource planning (ERP) systems or other software solutions commonly used by businesses. The project assumes the availability and easy deployment of hardware components such as IoT devices, sensors, and RFID tags without the need for extensive infrastructure modifications. The system is designed to be flexible and compatible with various industry-standard hardware, accommodating different business setups. Scalability is a key consideration, with the system being able to handle increasing data volumes and expanding inventory requirements, ensuring long-term viability and adaptability. Security and privacy are paramount, and the system assumes the implementation of robust measures to protect sensitive inventory information from unauthorized access or breaches. It also ensures compliance with relevant data protection regulations to safeguard the privacy of businesses and customers alike.

Main Part

The aim of this project is to develop a smart inventory management system that will allow businesses to efficiently manage their inventory and streamline their operations. The system will use technology like IoT (Internet of Things) to track and manage inventory in real-time.

The main objective of this project is to create a smart inventory management system that utilizes advanced technologies like IoT to effectively manage inventory and improve business operations. This system will provide real-time tracking and management of inventory, automate stock replenishment, and optimize inventory levels using forecasting and historical sales data. By implementing this system, businesses can experience benefits such as increased efficiency, cost savings, improved customer service, and better decision-making based on data insights. Ultimately, this project aims to help businesses enhance their inventory management practices, minimize waste, and operate more efficiently.

Upon the completion of the project to develop a smart inventory management system that leverages IoT technology to track and manage inventory in real-time, the following accomplishments could be achieved:

- The smart inventory management system would enable businesses to have real-time access to inventory data, leading to more efficient and accurate management of inventory levels. This

could lead to a reduction in inventory holding costs, improved order fulfilment, and increased customer satisfaction.

- The use of IoT technology to track and manage inventory in real-time would reduce the need for manual labour, minimizing the potential for human error in inventory management processes. This would free up time for employees to focus on other important tasks, leading to increased productivity and efficiency.

Project Objective

The project objective is to create a smart inventory management system that utilizes advanced technologies like IoT to effectively manage inventory and improve business operations. This system will provide real-time tracking and management of inventory, automate stock replenishment, and optimize inventory levels using forecasting and historical sales data.

The goal of the project is to improve the accuracy and efficiency of inventory management while reducing costs associated with manual inventory tracking and the potential for errors. By implementing an IoT-supported inventory management system, the SMART warehouse will be able to reduce waste, and enhance customer. This project provides that products are always in stock and readily available.

The Deliverables

The deliverables of the project are mainly under these topics.

- Project Plan is the first deliverable. This deliverable outlines the project scope, objectives, and timelines. It will include a detailed plan of action for completing the project, including milestones and deadlines.
- System Requirements Document is another important one. It outlines the financial and technical requirements of the smart inventory management system. It will include details about the technology and equipment needed to build the system.
- Design Document also outlines the architecture, user interface, and workflows of the system. This deliverable will include mock-ups, wireframes, and user stories to help visualize the system's functionality.
- Prototype is a working model of the system that demonstrates its basic functionality. It could have a limited set of features and be used for testing and validation.
- Final System: This deliverable is the fully functional and tested version of the smart inventory management system that meets all the requirements outlined in the System Requirements Document.
- Training Material is deliverable that provides materials for training staff on how to use the smart inventory management system.
- Maintenance Plan will maintain the smart inventory management system after it has been deployed. It includes details about how to resolve any issues that arise and how to update the system as needed.

Important Milestones

- Defining the project purpose and objectives.
- Identifying stakeholders and creating a communication plan.
- Preparation of flow chart.
- Identifying inventory management requirements.
- Identification of technical requirements and/or constraints.
- Program design and creation of architecture (UI/UX – Frontend).
- Development of software code and database schema.
- Managing unit testing and integration testing.
- Development of test scenarios.
- Performing user acceptance testing with stakeholders.
- Addressing defects or problems found during testing.
- Developing user training materials and organizing sessions.
- Deploy the inventory program to the production environment.
- Performing post-deployment testing and troubleshooting found issues.
- Establishing ongoing maintenance and support procedures.
- Ensuring ongoing support and handling user-reported issues.
- Monitoring system performance and making improvements as needed.

Technical requirements

The system seeks to give users an effective interface for managing inventory, and it will also give them a variety of alternatives through the many functionalities available. The user is informed when particular ingredient levels are low as a result of continual monitoring of ingredient levels based on consumption and inventory threshold levels. The system updates the inventory on the user's behalf, eliminating the need for the user to do it manually each time. The system calculates and forecasts how much will be used on particular days that the user (admin) has pre-defined; it also warns the user of an upcoming action to order ingredients before the particular day that the user has selected. Because the System performs it for the user, the user never has to bother about manually calculating the estimated usage of the ingredients. The System's straightforward user interface allows you to add, remove, or update recipes. Additionally, it includes operations like processing orders, changing processed orders, verifying threshold levels, adding a vendor for an ingredient, removing the vendor, and more.

- The system must be easy to use by both managers and chefs such that they do not need to read an extensive number of manuals.
- The system must be quickly accessible by both managers and chefs.
- The system must be intuitive and simple in the way it displays all relevant data and relationships.
- The menus of the system must be easily navigable by the users with buttons that are easy to understand.
- The System must give accurate inventory status to the user continuously. Any inaccuracies are taken care by the regular confirming of the actual levels with the levels displayed in the system.

- The System must successfully add any recipe, ingredients, vendors or special occasions given by the user and provide estimations and inventory status in relevance with the newly updated entities.
- The system must provide a password enabled login to the user to avoid any foreign entity changing the data in the system.
- The system should provide the user updates on completion of requested processes and if the requested processes fail, it should provide the user the reason for the failure.
- The system should not update the data in any database for any failed processes.

Limits, Assumptions, and Exclusions

There may be some limitations, assumptions, and exclusions associated with this project. For example, the project budget and timeline may be limited, and resources such as personnel and equipment may also be limited.

On the assumption side, it's important to assume that stakeholders will be supportive of the project and provide the necessary resources and support. Additionally, it's important to assume that the required technology, such as IoT sensors and gateways, will be readily available, and that the project team will have the necessary expertise to complete the project.

Finally, there are some exclusions to keep in mind. For example, the physical installation of the IoT sensors and gateways is not included in the project scope. Additionally, the project does not include the development of a custom ERP system for inventory management.

Stakeholders

The stakeholders of the project, especially in the warehouses, would be the suppliers. The information about the inventory, like inventory level, supply needs etc. , Is shared among the warehouse manager and supplier. Thanks to this intelligent communication, needed actions can be taken.

Customers

The customers of the project are owners of the goods, which are stored in the warehouse. Customers generally orders goods from the warehouses and the warehouse should be ready to serve them properly. Intelligent inventory management helps the warehouse manager and the company to satisfy the customer needs and services.

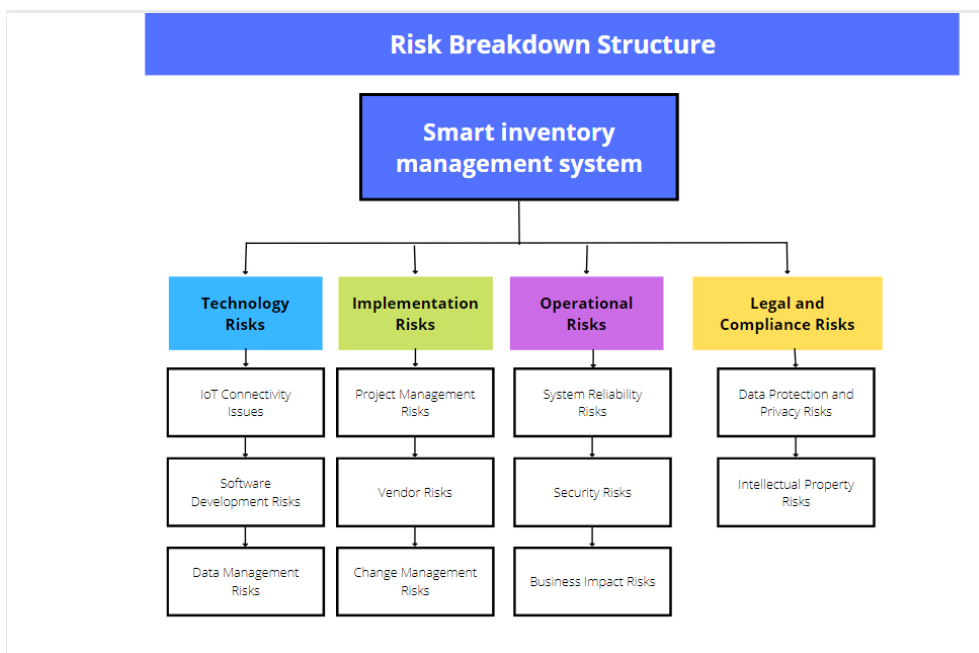
Responsibility Matrix

Task	Bora	M.Emin	Tuğçe	M.Enes	Serhan	Beyza
Project planning and scheduling		X	X			
Budget management and financial planning	X			X		
Stakeholder management		X	X			
Risk management				X	X	
Project reporting and documentation					X	
Identify stakeholders and their requirements						X
Conduct market research						X
Identify business processes and workflows				X		X
Define system requirements					X	
Develop use cases and scenarios	X					
System architecture design		X	X			
Database design	X	X	X			X
User interface design						
Network infrastructure design		X	X	X		X

Security design	X				X	
Hardware development and integration		X	X			
Software development	X			X		X
Integration of IoT sensors with software system	X					
Testing and validation of the system				X	X	
Deployment planning				X	X	
User training and support		X	X			
Maintenance and support	X				X	
Documentation and knowledge transfer				X		X

Table 1-Responsibility Matrix

Risk Breakdown Structure



Technology Risks

1.1. IoT Connectivity Issues- Unreliable or intermittent internet connectivity for IoT devices -Compatibility issues between different IoT devices and protocols - Data security and privacy concerns related to IoT devices.

1.2. Software Development Risks- Integration challenges between the inventory management software and IoT devices - Bugs or errors in the software that affect data accuracy or system functionality - Difficulty in scaling the software to handle large volumes of data.

1.3. Data Management Risks- Data quality issues, such as inaccurate or

incomplete data from IoT devices - Data synchronization problems between the inventory management system and IoT devices - Data loss or corruption due to system failures or cyber threats.

Implementation Risks

1.4. Project Management Risks- Lack of clear project objectives and scope - Inadequate resource allocation and project planning - Poor communication and coordination among project stakeholders

1.5. Vendor Risks- Dependence on external vendors for IoT devices, software, or services - Vendor delays, non-compliance with specifications, or product quality issues - Vendor lock-in or lack of vendor support in the long term

1.6. Change Management Risks- Resistance from employees or stakeholders to adopt new. inventory management processes - Training and skill gaps for employees to effectively use the new system - Disruption to ongoing business operations during the implementation phase.

Operational Risks

1.7. System Reliability Risks- Downtime or system failures affecting inventory management operations - Inadequate backup and recovery mechanisms for system and data - Insufficient system monitoring and maintenance practices

1.8. Security Risks - Unauthorized access or breaches of sensitive inventory data - Vulnerabilities in IoT devices or software leading to security breaches - Lack of proper access controls and authentication mechanisms

1.9. Business Impact Risks - Inaccurate inventory data leading to stockouts or overstocking - Inefficiencies in inventory management processes affecting overall business operations – Inability to meet customer demands due to system limitations or errors.

Legal and Compliance Risks

1.10. Data Protection and Privacy Risks - Compliance with data protection regulations (e.g., GDPR, CCPA) - Secure handling and storage of personally identifiable information (PII) - Obtaining necessary consent for collecting and using data from IoT devices.

4.2. Intellectual Property Risks - Infringement of patents or copyrights related to IoT technologies or software - Ensuring proper licensing and usage rights for third-party software or components - Protection of company's own intellectual property related to the inventory management system.

Risk Assessment Plan

Risk Identification

- Conduct brainstorming sessions and interviews with project stakeholders to identify potential risks related to the project.
- Analyze the project scope, objectives, and deliverables to

identify specific areas where risks may arise.

- Review industry best practices and case studies to identify common risks associated with implementing IoT-based inventory management systems.

Risk Analysis

- Assess the likelihood and impact of each identified risk. Use a scale (e.g., low, medium, high) to quantify the level of risk.
- Prioritize risks based on their potential impact on project objectives, schedule, budget, and overall success.
- Identify potential root causes and contributing factors for each risk to understand their underlying sources.

Risk Evaluation

- Determine the overall risk tolerance level for the project. Define acceptable levels of risk based on the project's objectives and stakeholders' expectations.
- Evaluate the identified risks against the established risk tolerance level to determine the significance of each risk.
- Classify risks as high, medium, or low based on their likelihood and impact.

Risk Response Planning

- Develop strategies and action plans to mitigate or address the identified risks.
- Identify risk mitigation measures, such as implementing redundant connectivity options for IoT devices or conducting regular security audits.
- Assign responsibilities and timelines for each risk response measure.
- Consider alternative strategies or contingency plans to minimize the impact of high-risk events.

Risk Monitoring and Control

- Establish a risk monitoring system to track the status of identified risks throughout the project lifecycle.
- Regularly review and update the risk assessment to account for new risks or changes in the project environment.
- Continuously monitor risk triggers and indicators to identify early warning signs of potential risks.
- Implement risk control measures and adjust response plans as necessary.

Communication and Reporting

- Communicate the risk assessment plan, including identified risks, mitigation measures, and risk tolerance levels, to

project stakeholders.

- Provide regular progress reports on risk management activities, including updates on risk mitigation measures, changes in risk status, and overall risk exposure.
- Ensure clear and effective communication channels are established to facilitate timely reporting of risks and issues.

Lessons Learned and Continuous Improvement

- Conduct post-project reviews to evaluate the effectiveness of the risk assessment plan and its associated strategies.
- Document lessons learned from the project, including successful risk mitigation measures and areas for improvement.
- Use the lessons learned to update and enhance future risk assessment plans and project management practices.

Our risk assessment plan might also involve the stakeholders, such as project managers, technical experts, and business representatives.

Risk Assessment Form

Risk Event	Likelihood	Impact	Detection Difficulty	When
IoT Connectivity Issues	3	5	2	Start-up
Project Management Risks	2	3	3	Installation
Security Risks	2	5	2	Postinstallation
Software Development Risks	5	4	3	Conversion

Figure 1- Risk assessment form

Project: Smart Inventory Management System

Risk Identification

1. **Risk Description:** IoT Connectivity Issues
2. **Likelihood:** High
3. **Impact:** Medium
4. **Risk Category:** Technology
5. **Risk Owner:** IT Department
6. **Risk Mitigation Strategy:** Implement redundant connectivity options for IoT devices, conduct thorough compatibility testing, and establish data backup mechanisms.
7. **Contingency Plan:** Establish a manual inventory management process

as a backup during connectivity issues.

8. **Actions Taken:** Conducted initial connectivity tests and identified potential solutions. Ongoing monitoring and evaluation of connectivity issues.

Risk Severity Matrix

Likelihood	5				Software Development Risks	
	4					
	3					IoT Issues
	2			Project Management Risks		Security Risks
	1					
		1	2	3	4	5
		Impact				

Figure 2-Risk severity matrix

Risk Response Plan

1. IoT Connectivity Issues

Risk Mitigation Strategy:

- Implement redundant connectivity options for IoT devices, such as multiple network providers or backup communication channels.
- Conduct thorough compatibility testing and ensure IoT devices are compatible with the selected protocols and network infrastructure.
- Establish data backup mechanisms to minimize the impact of connectivity disruptions.
- **Contingency Plan:** Establish a manual inventory management process as a backup during connectivity issues, allowing for continued operations with minimal disruption.
- **Risk Owner:** IT Department

2. Software Development Risks

Risk Mitigation Strategy:

- Conduct thorough integration testing between the inventory management software and IoT devices to identify and address any integration challenges early on.
- Implement software development best practices,

including code reviews, testing, and quality assurance processes, to minimize bugs and errors.

- Regularly monitor and update the software to address scalability and performance issues.

Contingency Plan: Maintain a list of alternative software solutions or vendors to consider in case major issues arise during development.

- **Risk Owner:** Development Team

3. Project Management Risks

Risk Mitigation Strategy

- Define clear project objectives, scope, and deliverables at the beginning of the project to ensure alignment and avoid scope creep.
- Allocate adequate resources, including personnel and budget, to the project to prevent delays and cost overruns.
- Establish effective communication channels and regular project meetings to facilitate coordination and decision-making among stakeholders.

Contingency Plan: Develop a comprehensive change management plan to address potential resistance to change and ensure smooth adoption of the new inventory management system.

- **Risk Owner:** Project Manager

4. Security Risks

Risk Mitigation Strategy:

- Implement robust security measures, including firewalls, encryption, and access controls, to protect sensitive inventory data.
- Regularly conduct security audits and vulnerability assessments to identify and address any weaknesses or vulnerabilities in the system.
- Provide ongoing training and awareness programs to employees regarding cybersecurity best practices to mitigate the risk of human errors.

Contingency Plan

- Develop an incident response plan to address any security breaches promptly, including procedures for investigation, containment, and recovery.
- **Risk Owner:** IT Security Team

Risk Response Matrix

Risk Event	Response	Contingency Plan	Trigger	Who is Responsible
IoT Connectivity Issues	Mitigate: Implement redundant connectivity options for IoT devices	Establish a manual inventory management process as a backup during connectivity issues, allowing for continued operations with minimal disruption.	Equipment fails	Tuğçe
Project Management Risks	Mitigate: Conduct thorough integration testing between the inventory management software and IoT devices	Maintain a list of alternative software solutions or vendors to consider in case major issues arise during development.	Still not solved in two hour	Beyza
Software Development Risks	Mitigate: Define clear project objectives, scope, and deliverables at the beginning of the project	Develop a comprehensive change management plan to address potential resistance to change and ensure smooth adoption of the new inventory management system	When Software fails	Serhan
Security Risks	Mitigate: Implement robust security measures, including firewalls, encryption, and access controls	Develop an incident response plan to address any security breaches promptly, including procedures for investigation, containment, and recovery.	When the Call comes from top management	Bora

Figure 3-Risk response matrix

Define Work

The goal of the project is to create a smart inventory management system that will help firms manage their inventories and streamline their processes. This leads to the creation of the WBS. Project planning, business analysis, requirements gathering, design and architecture, development and integration, deployment and support are the main components of the WBS.

Work packages serve as the sub-components of each of these parts. These job packages are noted and explained in the earlier assignments as well as in the examples below:

Project Preparation

- 1.1 Project planning and scheduling
- 1.2 Budget management and financial planning
- 1.3 Stakeholder management
- 1.4 Risk management
- 1.5 Project reporting and documentation

Business Analysis and Requirements Gathering

- 2.1 Identify stakeholders and their requirements

- 2.2 Conduct market research
- 2.3 Identify business processes and workflows
- 2.4 Define system requirements
- 2.5 Develop use cases and scenarios

Design and Architecture

- 3.1 System architecture design
- 3.2 Database design
- 3.3 User interface design
- 3.4 Network infrastructure design
- 3.5 Security design

Development and Integration

- 4.1 Hardware development and integration (IoT sensors, gateways)
- 4.2 Software development (Backend and Frontend)
- 4.3 Integration of IoT sensors with software system
- 4.4 Testing and validation of the system

Deployment and Support

- 5.1 Deployment planning
- 5.2 User training and support
- 5.3 Maintenance and support
- 5.4 Documentation and knowledge transfer

Identification of the to Complete a Work Package

The durations of the each work package is shown below:

Görev Adı	Duration	Start	Finish
WBS	90 days	Mon 03.04.23	Fri 04.08.23
Project Preparation	10 days	Mon 03.04.23	Fri 14.04.23
Project planning and scheduling	1 day	Mon 03.04.23	Mon 03.04.23
Budget management and financial planning	1 day	Tue 04.04.23	Tue 04.04.23
Stakeholder management	3 days	Wed 05.04.23	Fri 07.04.23
Risk management	3 days	Sat 08.04.23	Tue 11.04.23
Project reporting and documentation	2 days	Wed 12.04.23	Thu 13.04.23
Business Analysis and Requirements Gathering	10 days	Fri 14.04.23	Thu 27.04.23
Identify stakeholders and their requirements	2 days	Fri 14.04.23	Sat 15.04.23
Conduct market research	3 days	Tue 18.04.23	Thu 20.04.23
Identify business processes and workflows	2 days	Fri 21.04.23	Sat 22.04.23
Define system requirements	1 day	Mon 24.04.23	Mon 24.04.23
Develop use cases and scenarios	3 days	Tue 25.04.23	Thu 27.04.23
Design and Architecture	21 days	Fri 28.04.23	Fri 26.05.23
System architecture design	3 days	Fri 28.04.23	Tue 02.05.23
Database design	6 days	Wed 03.05.23	Wed 10.05.23
User interface design	3 days	Thu 11.05.23	Mon 15.05.23
Network infrastructure design	5 days	Tue 16.05.23	Sun 21.05.23
Security design	5 days	Mon 22.05.23	Fri 26.05.23

Development and Integration	40 days	Fri 26.05.23	Thu 20.07.23
Hardware development and integration (IoT sensors, gateways)	5 days	Fri 26.05.23	Thu 01.06.23
Software development (Backend and Frontend)	25 days	Tue 02.05.23	Mon 05.06.23
Integration of IoT sensors with software system	4 days	Thu 06.07.23	Tue 11.07.23
Testing and validation of the system	6 days	Wed 12.07.23	Wed 19.07.23
Deployment and Support	11 days	Thu 20.07.23	Thu 03.08.23
Deployment planning	3 days	Thu 20.07.23	Mon 24.07.23
User training and support	2 days	Tue 25.07.23	Wed 26.07.23
Maintenance and support	4 days	Thu 27.07.23	Tue 01.08.23
Documentation and knowledge transfer	2 days	Wed 02.08.23	Thu 03.08.23

Figure 4-Durations of the each work package

Identification of the Time-Phased Budget

The project's time-phased budget for finishing each major portion is displayed below. Depending on the requirements of the particular project, the time frame and financial allocation may change.

Project Preparation

Estimated Duration: 3 days.

Budget Allocation: \$10,000

Project planning and scheduling: \$3,000

Budget management and financial planning: \$2,000

Stakeholder management: \$2,000

Risk management: \$2,000

Miscellaneous expenses: \$1,000

Business Analysis and Requirements Gathering

Estimated Duration: 3 days

Budget Allocation: \$20,000

Identify stakeholders and their requirements\$5,000

Conduct market research: \$5,000

Identify business processes and workflows: \$4,000

Define system requirements: \$3,000

Develop use cases and scenarios: \$3,000

Design and Architecture

Estimated Duration: 17 days

Budget Allocation: \$30,000

System architecture design: \$8,000

Database design: \$6,000

User interface (UI) design: \$6,000

Network infrastructure design: \$5,000

Security design: \$5,000

Development and Integration

Estimated Duration: 25 days

Budget Allocation: \$60,000

Hardware development and integration (IoT sensors, gateways): \$15,000

Software development (backend and frontend): \$25,000

Integration of IoT sensors with software: \$10,000

Testing and validation of the system: \$10,000

Deployment and Support

Estimated Duration: 4 days

Budget Allocation: \$20,000

Deployment planning: \$5,000

User training and support: \$7,000

Maintenance and support: \$5,000

Documentation and knowledge transfer: \$3,000

These cost amounts are only estimates and may change based on the difficulty of the project, the size of the team, the cost of the resources, and other considerations.

Identification of the Resources

Project Preparation

Project Manager: Responsible for overall project planning, scheduling, and coordination.

Business Analyst: Assists in identifying stakeholders, conducting feasibility studies, and creating project charters.

Subject Matter Experts (SMEs): Provide expertise in relevant areas, such as inventory management or IoT technology.

Project Team Members: Assorted roles based on project requirements.

Business Analysis and Requirements Gathering

Business Analyst: Leads the requirements gathering process, conducts stakeholder interviews, and documents requirements.

Market Research Analyst: Conducts market research, industry analysis, and competitive analysis.

SMEs: Provide subject matter expertise in specific areas related to the business processes and requirements.

Design and Architecture

System Architect: Responsible for designing the overall system architecture, including hardware, software, and network infrastructure.

Database Designer: Designs the database structure and ensures data integrity and efficiency.

UI/UX Designer: Designs the user interface to ensure optimal user experience.

Security Specialist: Assesses security requirements and designs appropriate security measures for the system.

Development Team: Includes software developers, hardware engineers, and network specialists who implement the designs.

Development and Integration

Software Developers: Responsible for developing the backend and frontend of the inventory management system.

Hardware Engineers: Develop and integrate IoT sensors, gateways, and other hardware components.

Quality Assurance/Testers: Conduct testing and validation activities to ensure system functionality and reliability.

Integration Specialists: Ensure seamless integration between software components and IoT sensors.

Deployment and Support

Deployment Manager: Oversees the deployment planning and execution.

Trainers: Conduct user training sessions to ensure proper system usage.

Technical Support Team: Provides ongoing support and maintenance for the system.

Documentation Specialist: Creates and maintains project documentation and knowledge transfer materials.

Identification of a Group Members Responsibilities

Responsibility Matrix						
Task	Bora	M.Emin	Tuğçe	M.Enes	Serhan	Beyza
Project planning and scheduling		X	X			
Budget management and financial planning	X			X		
Stakeholder management		X	X			
Risk management				X	X	
Project reporting and documentation					X	
Identify stakeholders and their requirements						X
Conduct market research						X
Identify business processes and workflows				X		X
Define system requirements					X	
Develop use cases and scenarios	X					
System architecture design		X	X			
Database design	X	X	X			X
User interface design						
Network infrastructure design		X	X	X		X
Security design	X				X	
Hardware development and integration		X	X			
Software development	X			X		X
Integration of IoT sensors with software system	X					
Testing and validation of the system				X	X	
Deployment planning				X	X	
User training and support		X	X			
Maintenance and support	X				X	
Documentation and knowledge transfer				X		X

Table 2-Responsibility matrix

Identification of Monitoring Points for Measuring Progress

Project Preparation

Completion of the project charter and feasibility study.

Development of the project schedule and identification of key milestones.

Stakeholder engagement and alignment on project objectives.

Progress in assembling the project team and defining their roles and responsibilities.

Budget estimation and allocation.

Business Analysis and Requirements Gathering

Documentation of stakeholder requirements and their alignment with project objectives.

Conducting market research and analysis, including competitor analysis and industry trends.

Completion of business process analysis and documentation.

Validation and sign-off on functional and non-functional requirements.

Development of use cases, user stories, and process flow diagrams.

Design and Architecture

Completion of the system architecture design, including hardware, software, and network infrastructure.

Database design and validation of the data structure.

User interface (UI) design and validation of usability and user experience.

Security design and alignment with industry best practices.

Progress in the development of design artifacts and related documentation.

Development and Integration

Milestones achieved in software development, including backend and frontend implementation.

Progress in hardware development and integration of IoT sensors.

Successful integration of IoT sensors with the software system.

Completion of unit testing, system testing, and user acceptance testing.

Identification and resolution of defects or issues discovered during testing.

Deployment and Support

Completion of the deployment plan and successful system deployment.

User training conducted and feedback received from training sessions.

Post-deployment support provided and response time for resolving user inquiries or issues.

Maintenance activities carried out as per the defined schedule.

Regular updates to project documentation and knowledge transfer materials.

These monitoring locations serve as milestones to gauge how each main component is progressing.

They may be used by project managers to keep tabs on the completion of crucial activities, verify deliverables, and make sure the project is on schedule to meet its goals. Depending on the demands of the particular project and the expectations of the stakeholders, adjustments may be made as necessary.

What is Project Monitoring System and what is it used for?

A Project Monitoring System is a structured approach or set of processes and tools used to track, assess, and manage the progress and performance of a project. It involves collecting data, analyzing it, and generating reports to provide insights into the project's status, identify potential issues or risks, and make informed decisions.

Tracking Progress: It helps monitor the progress of tasks, activities, and milestones in a project. This allows project managers to determine if the project is on schedule and identify any delays or bottlenecks that may arise.

Data Collection: The system collects relevant data about the project, including task progress, resource utilization, budget expenditures, quality metrics, and any other parameters necessary to evaluate project performance.

Identifying Issues and Risks: A Project Monitoring System helps identify and track potential issues or risks that may affect the project's success. By monitoring these risks, project managers can develop mitigation strategies and take proactive actions to minimize their impact.

Reporting and Communication: The system generates reports and provides data visualizations that offer a clear overview of the project's status and performance. These reports facilitate effective communication among project stakeholders, ensuring everyone is informed and aligned.

How can we implement this system for our Smart Inventory Management Project?

The project monitoring system for our smart inventory management project will help us track the progress of our project, identify any bottlenecks or issues, and ensure that the project is on track towards successful completion. The system will collect relevant data and generate reports to provide insights into the project's status, performance, and potential areas of improvement.

Data to Collect

Project Milestones: Record the planned milestones and their corresponding deadlines.

Task Progress: Track the progress of each task in the project, including start dates, end dates, and completion status.

Resource Allocation: Monitor the allocation of resources (e.g., team members, equipment, budget) to different tasks.

Inventory Data: Collect real-time data from IoT devices to track inventory levels, including current stock quantities, incoming shipments, outgoing orders, and any discrepancies.

Issues and Risks: Log any issues or risks that arise during the project, along with their impact and proposed mitigation plans.

Communication Logs: Maintain a record of project-related communications, including meetings, emails, and discussions.

Reports to Generate

Project Overview: Provide a high-level summary of the project's status, including milestones achieved, pending milestones, and overall progress.

Task Progress Report: Display the progress of individual tasks, highlighting completed tasks, tasks in progress, and overdue tasks.

Resource Utilization Report: Show the allocation of resources across different tasks, identifying any imbalances or underutilized resources.

Inventory Status Report: Present real-time inventory data, including current stock levels, incoming and outgoing orders, and any discrepancies or stockouts.

Issues and Risks Report: Highlight identified issues and risks, their impact on the project, and the progress of mitigation efforts.

f. Communication Log Report: Summarize project-related communications, providing a timeline of important discussions and decisions.

Performance Measurement

Amount of Completed Work: The ratio of the Completed part of the project to the whole project is 85% on 26.05.2023.

Time Spent: We have completed 86% of our project on 26.05.2023, which corresponds to 78 days in the allotted time. Our project complies with the time allotted as of today, and 86% by using the 2 day period we have advanced for the solution of the risks that may arise during the project construction time. It has been reached.

Budget: Today, 85% of our project has been completed, we have used 60% of the budget allocated for the project. In the project budget planning, 30% of the budget was reserved for putting our software project live. That is, for the last part of our project before it is completed.

Although we go about in parallel with our budget planning, we anticipate that it would be better to cut expenditures a little.

Cost Performance: We do not foresee an increase in money from the budget allocated in terms of cost performance. It is approximately parallel to the budget plan currently planned.

Scope Changes: At the request of our customer, there was a request that was not included in the project plan schedule and was added later. Our customer previously requested the addition of employees to the program while only requesting an inventory software program for their products. Although this made us think about the risk of extension of the project at first, the design of the product addition list to the database and since it is extremely similar to the frontend, it has been successfully completed by our employees without any extra workload and without disrupting the project.

Quality standards

The main quality standard of a company's inventory project for us is security. Any security vulnerability may cause the stock to be deducted from the stock. This may cause financial damage to the company. The security steps of our project have not been completed yet. In order to be 100% sure of the security of our software, we need to complete our project and take it live and perform the necessary security tests.

Management of Risks

1-One of our current risks is exceeding the project budget. In order to prevent this, the planned budget transfer was followed step by step and unnecessary expenditures were prevented.

2- When there is an update to our software programs, the malfunction of the old version is one of the risks that we encounter and manage successfully. One of our team has already determined the risks that may arise because this risk has been foreseen in advance. Since we know where to look for a solution when we encounter a problem, it is easy to adapt to the new version.

PROJECT TASKS

	Task	Completed Yes/No
1	All project deliverables have been completed	No
2	Documents are stored in full	Yes
3	Budget Used Correctly	Yes
4	Parallel to the project calendar planning	Yes
5	Brainstorming among the team	Yes
6	Agreed on the terms of the contract with the customer	Yes
7	Stakeholders informed about project progress	Yes
8	Evaluate the good and bad aspects of competitors	Yes
9	The latest version of the software has been compiled	Yes
10	Necessary application instructions for the software project were prepared	Yes
11	All necessary safety tests have been carried out	No
12	Veri Data security ensured	yes
13	Customer satisfaction and feedback were evaluated	No
14	Project documents archived	Yes
15	Problems that may arise when the project goes live were shared with stakeholders	Yes

Table 3-Project tasks

Performance of the Final Project

Name	Start	Finish	Actual Start	Actual Finish	Constraint Date	Created	Deadline	Duration	% Complete
Project SON	Mon 3.04.23	Fri 4.08.23	Mon 3.04.23	NA	NA	Sat 1.04.23	NA	90 days	86%

Table 4-WBS Table

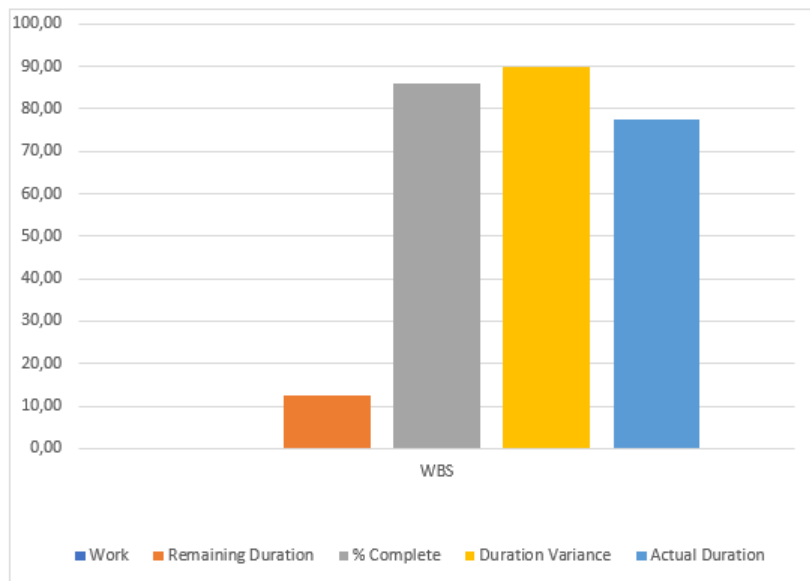


Figure 5-WBS Chart

As can be seen from the top, 86% of the tasks are finished. This shows that there are 14 days left out of a total of 90 days for the project to be finalized.

You can see the charts and tables from the MS Project File recent reports section.

PMI's Organizational Project Maturity Model

Our project aims to develop a smart inventory management system using IoT technology to track and manage inventory in real-time. The project objectives include improving inventory management efficiency, automating stock replenishment, and optimizing inventory levels based on forecasting and historical sales data. The deliverables of the project include a project plan, system requirements document, design document, prototype, final system, training material, and maintenance plan.

According to PMI's Organizational Project Maturity Model (OPM3), we set out the main factors to evaluate the maturity level of our final project efforts, such as project planning, stakeholder management, risk management, resource management, and overall project execution. We discussed through these main factors to understand our maturity level and how can we improve our standards of project maturity.

Project Planning: Our project demonstrates a clear understanding of the project scope, objectives, and deliverables. Key milestones and deadlines are identified, and there is a time-phased budget allocated for each work package. This indicates a certain level of maturity in project planning.

Stakeholder Management: The stakeholders of the project are identified, including suppliers and customers. However, there is limited information on how stakeholder requirements are managed and addressed throughout the project. We discussed about what else we could have done to improve the maturity model and we decided that More details about stakeholder engagement and communication plans would be needed to assess the maturity level.

Risk Management: A risk breakdown structure is provided, identifying technology risks and software development risks. However, there is no information on how these risks are assessed, mitigated, and monitored throughout the project. To move up our maturity level in terms of risk management, we could set out more comprehensive risk management approach.

Resource Management: Resources required for completing the work packages are identified throughout our project, including project manager, business analyst, subject matter experts, development team, trainers, and technical support team. This indicates a basic fundamental level of resource management.

Overall Project Execution: The project includes various work packages covering project preparation, business analysis, design and architecture, development and integration, deployment, and support. The milestones and monitoring points provide a basis for tracking progress. However, according to our discussions as a project group, the information provided does not give a complete picture of the project execution and control processes and this situation makes it difficult to assess the maturity level in this area.

Based on our evaluation, our entire project efforts demonstrate some level of maturity in project planning and a basic understanding of resource management. However, we can add more comprehensive information and details on stakeholder management, risk management, and overall project execution to assess the maturity level accurately according to PMI's OPM3.

Identification of Work Packages Suitable for Outsourcing

Network Management and Database

Infrastructure management usually requires constant maintenance, updating and monitoring. This requires special hardware, expert personnel, and continuous training costs. Our company wants to focus on its core business and strategic goals instead of dealing with infrastructure management and data storage. It provides another company that we have contracted for, but still our data is stored on our local servers (Monica). Outsourcing network management and data storage not only allows us to focus more on our primary business and be more successful, but also less costly to manage.

Software Testing and Quality Assurance

We use a contracted company as an outsource to detect security vulnerabilities of our software and take protection measures. The test company plans how the security test will be performed. Usually, test scenarios are created using predefined scenarios and attack vectors. In fact, the process is when our testing company attacks our firewall and tries to break it down. The vulnerabilities are a bit

difficult for the company that wrote the software to notice. That's why we need another view on this, so we outsource. In case the testing company breaks our firewall, the company notifies us of our vulnerabilities. Our company also starts improvement studies in this regard.

Support and Maintenance Services

We outsource our contracted company that provides technical support, troubleshooting and error correction and user support services to our customers. This company provides services to our customers such as answering questions about software usage and solving technical problems. It's because it's less costly.

Outsourcing Plan for Our Smart Inventory Management System

Define Project Scope: Clearly define the objectives, functionalities, and requirements of the smart inventory management system. This will help in identifying the specific tasks that can be outsourced.

Identify Outsourcing Partners: Research and identify potential outsourcing partners who specialize in software development, IoT integration, and inventory management systems. Look for companies with a proven track record in similar projects and positive client testimonials.

Request for Proposals (RFPs): Prepare a detailed RFP document outlining the project requirements, deliverables, timelines, and budget. Share the RFP with the selected outsourcing partners and ask them to submit their proposals.

Evaluation and Selection: Evaluate the received proposals based on factors such as expertise, experience, cost, timeline, and quality. Select the outsourcing partner that best aligns with your project requirements and business goals.

Contracts and Agreements: Once the outsourcing partner is selected, negotiate, and finalize the contractual terms and conditions. Ensure that the agreement includes confidentiality clauses, intellectual property rights, project milestones, and performance metrics.

Project Management: Assign a dedicated project manager from your team to oversee the outsourcing project. Establish regular communication channels and project management tools to facilitate effective collaboration and monitoring.

Progress Tracking: Implement a robust progress tracking mechanism to monitor the outsourced tasks. Regularly review the milestones, deliverables, and timelines to ensure the project stays on track.

Quality Assurance: Define quality standards and conduct regular quality checks at different stages of the project. Collaborate closely with the outsourcing partner to address any quality concerns promptly.

Communication and Reporting: Maintain open and transparent communication with the outsourcing partner throughout the project. Schedule regular progress meetings and require detailed progress reports to stay updated on the project status.

Integration and Testing: Collaborate closely with the outsourcing partner to integrate the developed system with the required IoT technologies and conduct comprehensive testing to ensure seamless functionality.

Training and Support: Plan for training sessions to familiarize the internal team with the smart inventory management system. Additionally, establish a support mechanism to address any issues or questions that may arise post-implementation.

Risks specific to this outsourcing plan:

Communication and Collaboration Challenges: Differences in time zones, language barriers, and cultural nuances can pose challenges in effective communication and collaboration between your team and the outsourcing partner. Establish clear communication channels, use collaboration tools, and promote open and regular communication to mitigate these risks.

Quality Control: Outsourcing a significant part of the project introduces the risk of differences in quality standards or misunderstandings between your expectations and the outsourcing partner's deliverables. Implement a strong quality assurance process, conduct regular reviews, and provide detailed feedback to address any quality concerns promptly.

Security and Data Privacy: The use of IoT technology involves handling sensitive data related to inventory and operations. Ensure the outsourcing partner has robust security measures in place to protect data integrity and privacy. Implement appropriate security protocols and conduct regular audits to mitigate the risk of data breaches.

Project Partnering Charter

Project Name: [Inventory Software Program]

Date: [29.03.2023]

1. Project Overview: [The purpose of the software will enable efficient tracking of inventory levels, facilitate inventory replenishment, and provide accurate and real-time reporting capabilities. This will increase overall operating efficiency, decrease stockouts and overstocks, and improve inventory control.]
2. Project Partners: [Software X is in charge of creating firewalls with the teams they provided in this project. The Hexa company will act as a distributor for this project and future projects.]
3. Project Objectives: [Develop a user-friendly inventory software program that meets the specific requirements of our organization. To ensure precise stock management and prompt replenishment, enable automated inventory level tracking. Create thorough reports and analytics to reveal information about the performance of your inventory, including turnover rates, stock accuracy, and carrying costs.]
4. Project Deliverables: [User documentation and training materials. Successful deployment of the software in the organization. Training sessions conducted for end-users. Ongoing support and maintenance plan. The project will last from 03.04.2023 until 04.08.2023. The timeline will be further detailed in the project schedule.]
5. Financial Considerations: Initial Cost: The planned cost of acquiring the program. License Fees, Implementation Cost, Hardware and Infrastructure Cost, Configuration and Customization Costs. Subscription and Maintenance Fees: Since our program is offered as a subscription, it includes the annual/monthly subscription and maintenance fees of the program. ROI: Potential return on investment. Scalability and Growth: The cost required to

expand the capabilities of our program as it grows, or to adapt to increased data volume, number of users, workload. Integration and Compliance: The cost of integrating into existing systems such as accounting, sales systems. Training and Support: The cost of training required to enable staff to use the program effectively. Maintenance and Upgrades: Continuing maintenance cost.]

6. Intellectual Property: Copyright; Our source code, user interface design, software and other creative elements made by our company belong to our company. Patents; Our software is intellectually and de facto owned by us. Intellectual Property Transfer Agreement; Defines the transfer and ownership of intellectual property from individuals to companies. License Agreement; It determines the use and integration conditions of the third party ideas used in our software. Confidentiality Agreement; It prevents the disclosure of confidential information shared during the project to unauthorized persons.]
7. Performance Evaluation: [The project was carried to the end on time and in accordance with the determined budget, as determined beforehand. So we can attribute the performance of the project successfully.]
8. Term and Termination: [In this project, a partnership agreement has been signed with our partners with whom we cooperate only within the scope of this project, and if the project is canceled for some reason, the partnership agreement will be broken in this way.]
9. Signatories: [In this contract, every detail and determined rules on behalf of the project have been accepted.]

By signing this Project Partnering Charter, the partners agree to collaborate effectively, communicate transparently, and work towards the successful completion of the project.

Signature: _____ Date: [29.03.2023] [Us]

Signature: _____ Date: [29.03.2023] [Software X]

Signature: _____ Date: [29.03.2023] [Hexa]

Conclusion

The software package offers a centralized database where companies can store and arrange crucial inventory data, including information about products, stock levels, and supplier specifics. This guarantees real-time visibility into inventory levels and empowers companies to choose wisely when it comes to buying, restocking, and order fulfillment.

The software application also makes it possible to follow stock movements automatically, removing the need for manual data entry and lowering the risk of human error. Businesses can

set up automatic reorder points and notifications thanks to this feature, which guarantees prompt replacement and avoids stockouts and overstocking.

The software also has strong reporting and analytics features, and it can generate in-depth analyses on important inventory parameters including turnover rates, carrying costs, and demand trends. With the use of these analytics, firms can spot trends, produce precise projections, and adjust inventory levels to satisfy client demand while reducing holding costs.

In conclusion, the inventory software program we have examined is a powerful tool that streamlines and optimizes inventory management processes for businesses. It offers a range of features and functionalities designed to enhance efficiency, accuracy, and cost-effectiveness.

Appendices

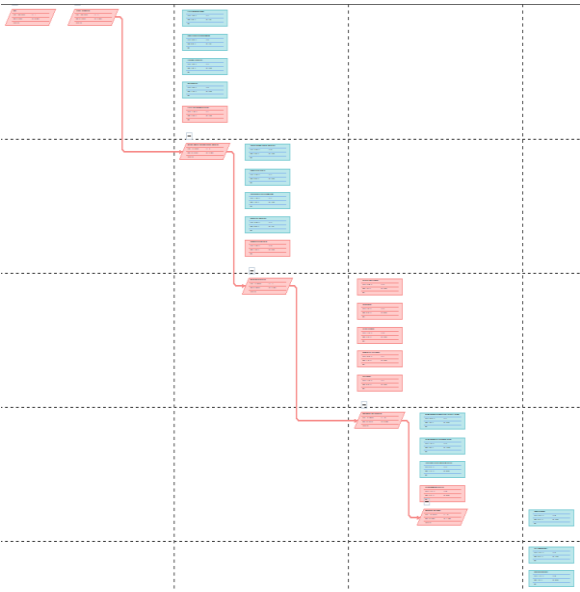


Figure 6-MS Project Part-1

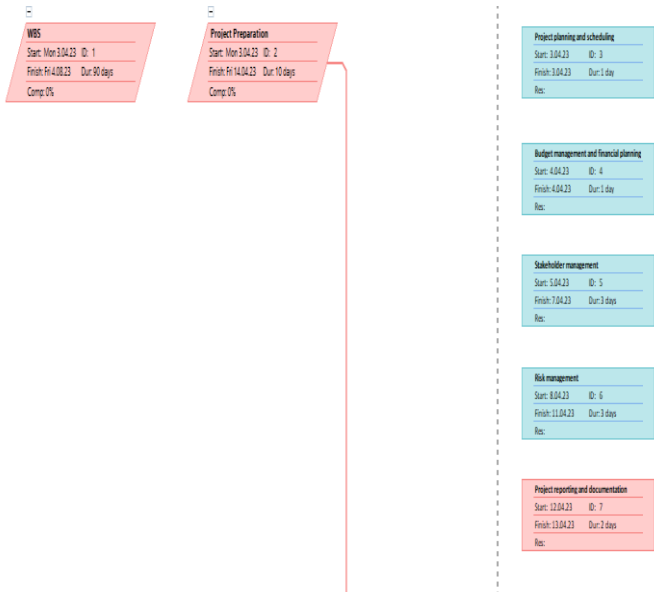


Figure 7-MS Project Part-2-Project preparations

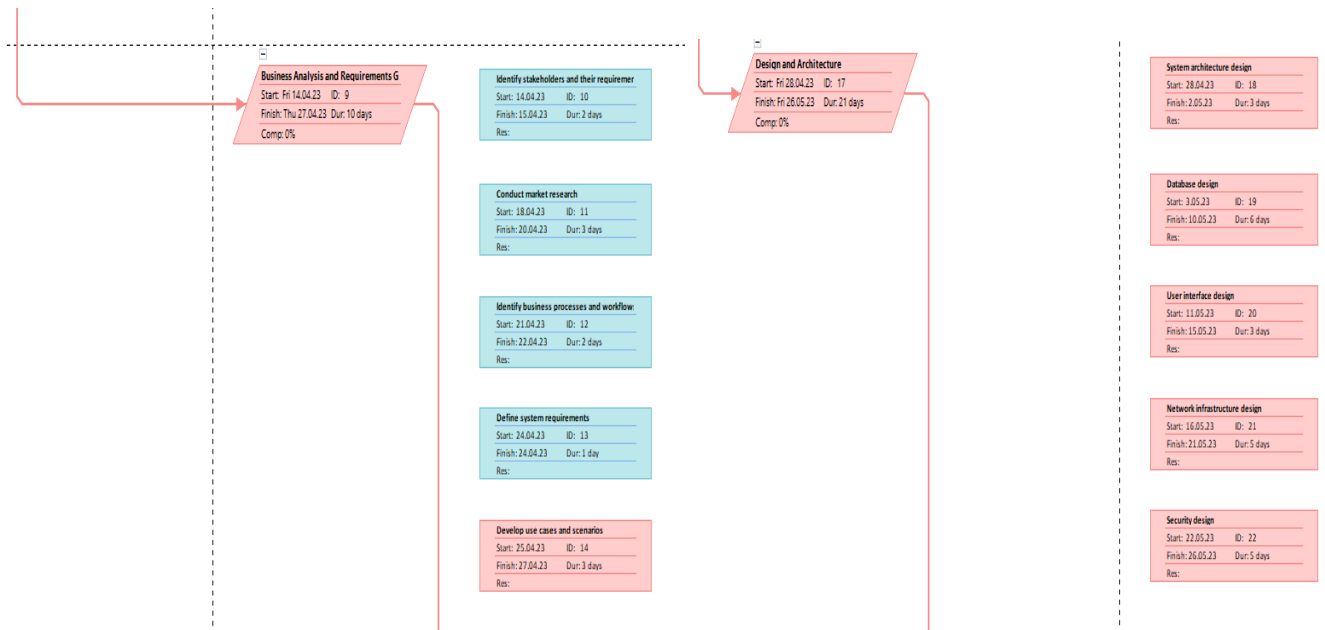


Figure 8-MS Project Part-3-Business Analysis and Requirements

Figure 9-MS Project Part-4-Design and Architecture

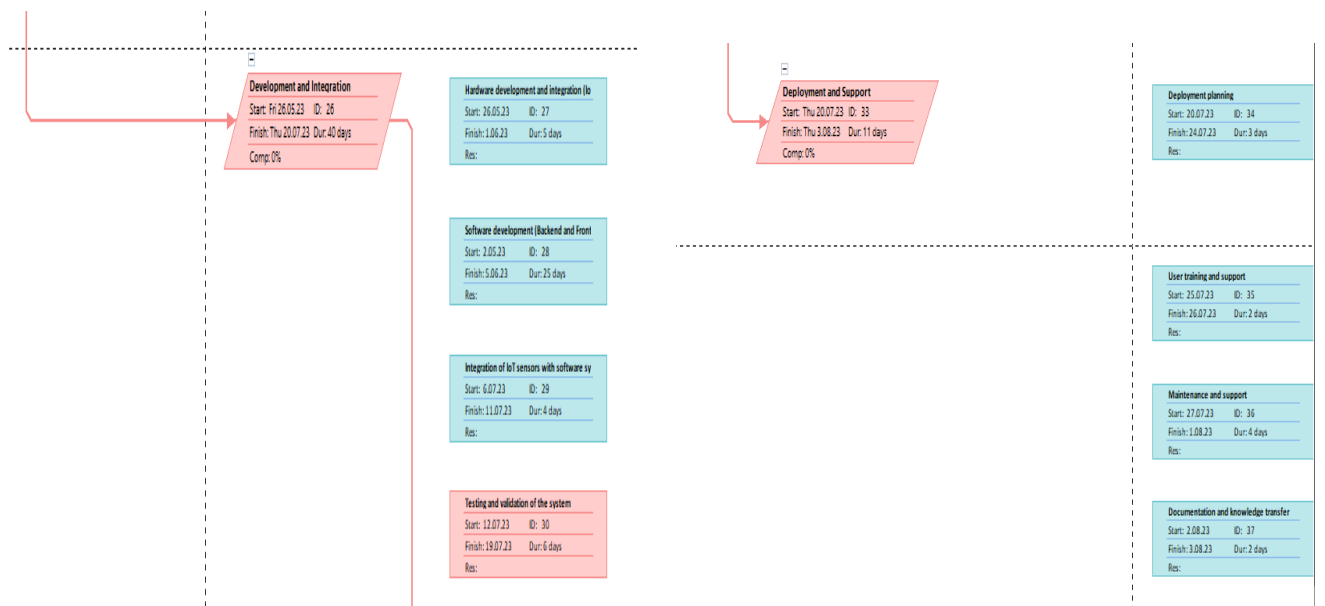


Figure 10-MS Project Part-5-Development and integrations

Figure 11-MS Project Part-6-Deployment and support

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